Estimating Cross-Lingual Tokenization Fragmentation Risk: A Frequency-Based Uniqueness Score for Geographic Entities

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Abstract

Large language models often penalize multilingual users through hidden 'tokenizer taxes' caused by subword fragmentation. Geographic proper nouns offer a controlled stress case for this distributional asymmetry, as they are standardized, cross-linguistically available entities whose orthographic forms frequently mismatch tokenizer training distributions. We introduce the Cross-Linguistic Orthographic Uniqueness Score (CLOUS), a character-level metric based on Shannon surprisal that quantifies the orthographic rarity of localized country names. We evaluate CLOUS against tokenization fertility for six production tokenizers spanning three algorithm families across eleven languages and seven script families. Using Spearman rank correlation with Benjamini-Hochberg correction, CLOUS shows significant associative strength with fertility in 61 of 65 valid language-tokenizer combinations. As a screening tool for high-fragmentation entities, CLOUS achieves ROC-AUC = 0.850 and F1 = 0.741, correctly identifying approximately three-quarters of the highest-fragmentation entities from orthographic properties alone. We further show that CLOUS correlates with fertility more strongly than two naive length-based baselines (raw byte length and character count) particularly for non-Latin scripts, suggesting that character-level surprisal captures information beyond string length. Script family appears to be the primary driver of CLOUS-fertility correlation strength, with effect sizes ranging from modest for Latin-script languages (ρ ≈ 0.17–0.43) to large for non-Latin scripts (ρ up to 0.686 for Japanese). These findings suggest that CLOUS may be useful as a lightweight pre-tokenization diagnostic for multilingual NLP pipelines. The code, datasets, and complete evaluation pipeline are publicly available at https://github.com/ric-rivera/CLOUS-tokenization-metrics.

Keywords: subword tokenization, tokenization fertility, cross-lingual NLP, geographic named entities, character-level surprisal, multilingual language models, orthographic complexity

1. Introduction

Subword tokenization is now a standard preprocessing step in large language model (LLM) development. Byte-Pair Encoding (BPE), WordPiece, and SentencePiece each learn to segment text by identifying co-occurring character sequences in large training corpora. The result is a vocabulary in which frequently observed character combinations receive compact token representations, while rare combinations are decomposed into shorter units or, at the extreme, individual bytes.

This decomposition carries practical costs. Longer token sequences increase the computational cost of self-attention quadratically, and fragmented subword representations have been shown to carry less semantic information than holistic representations of the same string (Rust et al., 2021; Ahia et al., 2023). The degree of fragmentation a string incurs is therefore not arbitrary: it is a function of how well the string's character distribution is represented in the tokenizer's learned vocabulary.

Geographic proper nouns offer a controlled test case for this phenomenon. Country names are standardized, ISO-registered entities with official localizations across every written human language, providing a consistent cross-lingual corpus for tokenization experiments. A name like France, composed of high-frequency Latin characters, tokenizes efficiently across Latin-script tokenizers. A name like Djibouti, whose character sequences are underrepresented in most training corpora, fragments more severely. This disparity follows predictably from the orthographic properties of the input string relative to the tokenizer's training distribution.

Existing work documents cross-lingual fertility disparities at the language level (Rust et al., 2021; Ahia et al., 2023) but does not provide a metric for estimating fragmentation risk at the entity level before model training or inference. Simple length-based heuristics, such as raw byte count or character count, might seem to approximate fragmentation risk, but they conflate string length with orthographic complexity and do not capture the distributional mismatch between an entity's character composition and a tokenizer's learned vocabulary.

This paper introduces CLOUS (Cross-Linguistic Orthographic Uniqueness Score) to address this gap. CLOUS computes the mean per-character surprisal of a geographic entity name relative to a language-specific character frequency distribution, yielding a scalar score that can be computed without model inference. The paper makes four contributions:

- We define CLOUS formally, specify the normalization and smoothing choices required for consistent cross-lingual computation, and demonstrate its derivation from Shannon information theory.
- We construct a reproducible evaluation pipeline benchmarking CLOUS against tokenization fertility for 249 country names across six production tokenizers and eleven languages.
- We show that CLOUS exhibits stronger correlational alignment with fertility than two naive length-based baselines, particularly for non-Latin scripts, providing empirical justification for the metric's design.
- We release the full pipeline as an open-source repository[1] to support extension to additional languages, tokenizers, or entity types.

2. Related Work

2.1 Subword Tokenization Algorithms

The three dominant tokenization paradigms in contemporary NLP differ in how they construct and apply vocabulary. Byte-Pair Encoding, introduced by Sennrich et al. (2016) for neural machine translation, iteratively merges the most frequent adjacent byte pairs in a training corpus until a target vocabulary size is reached. WordPiece, introduced by Schuster and Nakamura (2012) and used in BERT-family models, selects merges that maximize training data likelihood. Unigram Language Model tokenization as implemented in SentencePiece (Kudo and Richardson, 2018), used by XLM-RoBERTa and Gemma, prunes a large initial vocabulary by removing entries whose removal produces the smallest decrease in corpus likelihood. All three approaches derive token boundaries from corpus frequency statistics, making

[1] https://github.com/ric-rivera/CLOUS-tokenization-metrics

fragmentation severity a function of how well the input string's characters are represented in the training distribution.

2.2 Tokenization Fertility and Cross-Lingual Inequity

Fertility, defined as the ratio of output tokens to input characters or words, was formalized as a tokenization evaluation metric by Rust et al. (2021). Subsequent work has documented substantial fertility disparities across languages: non-Latin and lower-resource scripts consistently require more tokens to represent equivalent content, a phenomenon described by Rust et al. (2021) as a tokenization tax. Ahia et al. (2023) extended this analysis to commercial LLM APIs, demonstrating that the tax translates into higher per-inference costs for non-English users.

Research on vocabulary scaling reveals that larger vocabularies do not uniformly resolve cross-lingual fertility inequity. Zhang et al. (2024) found that the GPT-4o tokenizer's 200,000-token vocabulary introduced hallucination artifacts for under-trained long tokens in Chinese. Frick et al. (2024) evaluated tokenizer-free alternatives and found that large BPE vocabularies fail to close performance gaps for scripts orthographically distant from English. The present study complements this line of work by asking whether fragmentation risk for specific named entities can be estimated from a simple character-level metric before training or deployment.

While recent work documents significant few-shot performance degradation due to fertility-driven fragmentation (Ovcharov, 2026), these studies typically rely on post-hoc measurement. Our work provides a predictive diagnostic that allows developers to assess this risk for specific entities prior to model interaction.

2.3 Geographic Named Entity Processing

Named entity recognition for geographic locations has been widely studied (Tjong Kim Sang and De Meulder, 2003; Lample et al., 2016). However, the tokenization-level properties of geographic names have received less systematic attention. Beinborn et al. (2013) examined cross-lingual cognate detection without addressing tokenization implications. We are not aware of prior work proposing a corpus-frequency-based metric specifically designed to estimate geographic entity tokenization fragmentation across multiple languages and tokenizer families.

2.4 Information-Theoretic Approaches to Linguistic Complexity

The application of Shannon information theory (Shannon, 1951) to linguistic measurement includes work on syntactic surprisal in psycholinguistics (Hale, 2001; Levy, 2008), cross-linguistic comparisons of script efficiency (Moran and Cysouw, 2018), and reading time prediction (Smith and Levy, 2013). CLOUS is formally analogous to per-character cross-entropy adapted to entity-level characterization. The novelty is the application of this framework to tokenization behavior for a standardized, cross-linguistically available entity class, and the use of this metric for comparison against simpler length-based baselines.

2.5 How CLOUS Differs from Existing Information-Theoretic Measures

CLOUS is formally related to per-character cross-entropy but differs from prior information-theoretic linguistic measures in three respects. First, it operates at the entity level rather than the corpus or word level, enabling entity-specific fragmentation estimates that are not possible with language-level fertility measures.

Second, it uses language-specific character frequency distributions as the reference model, making it inherently cross-lingual and directly comparable across scripts without requiring parallel corpora or aligned annotations. Third, its explicit purpose is diagnostic: to serve as a pre-inference screening tool for tokenization fragmentation risk rather than a measure of linguistic complexity per se. The contribution of this work is not the surprisal calculation itself, which follows directly from Shannon (1951), but the application of this framework to geographic entity tokenization across a controlled, standardized cross-lingual corpus, and the demonstration that it outperforms naive length-based proxies while achieving ROC-AUC = 0.850 as a high-risk entity classifier.

3. Methodology

3.1 Dataset: Unicode CLDR Geographic Localizations

Country names were sourced from the Unicode Common Locale Data Repository (CLDR), a widely used resource for geographic name localization. We extracted territories.json files from the cldr-localenames-full package for each target locale. Country codes were filtered against the ISO 3166-1 alpha-2 whitelist using the pycountry library, excluding macro-region codes (EU, UN) and unrecognized territories. This yielded 249 valid country entries with complete localization coverage across all eleven languages. All strings were normalized to Unicode NFC form at ingestion to ensure that composed characters are not artificially decomposed into base-plus-combining-mark pairs.

3.2 Character Frequency Profiles

Language-specific character frequency profiles were constructed from the Leipzig Corpora Collection (Goldhahn et al., 2012), a multilingual corpus built from news and web text with consistent preprocessing across languages. We used 1-million-word corpora for all target languages. Character frequencies were derived from the word-frequency files by summing character occurrences weighted by word frequency. Table 1 summarizes the language matrix.

Code	Language	Script Family	Corpus Size
en	English	Latin	1M words
es	Spanish	Latin	1M words
fr	French	Latin	1M words
tr	Turkish	Latin (extended)	1M words
ru	Russian	Cyrillic	1M words
ar	Arabic	Arabic	1M words
hi	Hindi	Devanagari	1M words
ta	Tamil	Tamil	1M words
ko	Korean	Hangul	1M words
zh	Mandarin	Logographic	1M words
ja	Japanese	Mixed (Hiragana/Katakana/Kanji)	1M words

Table 1. Language matrix and corpus sources. All corpora normalized to Unicode NFC before frequency computation.

3.3 The CLOUS Formula

CLOUS is defined as the mean per-character surprisal of the NFC-normalized entity name, computed in bits:

$$\mathbf{CLOUS = -(1/N) \sum_{i=1}^{N} \log_2(\max(f(c_i), \varepsilon))}$$

where N is the number of characters in the normalized string, $f(c_i)$ is the relative frequency of character c_i in the target language corpus, and ε is a minimum frequency floor (default 1×10^{-7}) applied via max() to prevent undefined values for characters absent from the corpus. Higher CLOUS scores indicate that the entity's character composition is, on average, less typical of the target language's written form.

3.4 Baseline Metrics

To evaluate whether CLOUS captures information beyond string length, we define two baselines computed for each entity-language pair:

- Byte Length (BL): the length of the NFC-normalized string encoded in UTF-8, in bytes. This reflects both string length and script encoding cost (e.g., Arabic and CJK characters occupy 2–3 bytes in UTF-8).
- Character Count (CC): the number of Unicode code points in the NFC-normalized string. This reflects string length independent of encoding cost.

Spearman correlation between each baseline and fertility is reported alongside CLOUS correlations in Table 4. If CLOUS does not substantially outperform these baselines, the information-theoretic design of the metric would require further justification.

3.5 Tokenizer Selection

Six production tokenizers were selected to represent variation across algorithm families and vocabulary sizes. Table 2 lists the tokenizer registry.

ID	Model	Algorithm	Vocabulary Size
gpt4o	GPT-4o (o200k_base)	BPE	200,000
gpt4	GPT-4 (cl100k_base)	BPE	100,256
llama3	Llama 3.1-8B	BPE	128,000
gemma	Gemma-7B	SentencePiece	256,000
xlmr	XLM-RoBERTa	SentencePiece	250,000
mbert	mBERT (multilingual-cased)	WordPiece	110,000

Table 2. Tokenizer registry. Six tokenizers spanning three algorithm families and vocabulary sizes from 110,000 to 256,000 tokens.

3.6 Statistical Analysis

Tokenization fragmentation was operationalized using fertility (Rust et al., 2021): the number of output tokens divided by the number of input characters. A fertility score of 1.0 indicates one token per character; values above 1.0 indicate fragmentation. Spearman rank correlation (ρ) was used as the primary measure of associative strength, as it makes no distributional assumptions about CLOUS, baseline, or fertility scores. Benjamini-Hochberg correction was applied to all Spearman p-values to control the false discovery rate across 66 tests (one per language-tokenizer combination). Results are reported at the corrected α = 0.05 threshold. One combination (mBERT × zh-Hans) produced constant fertility values due to mBERT's character-level Chinese tokenization strategy, making Spearman correlation undefined; this combination was excluded, yielding 65 valid tests (see Appendix B).

4. Results

4.1 Associative Strength of CLOUS Across Languages

CLOUS was significantly correlated with tokenization fertility in 61 of 65 valid language-tokenizer combinations after Benjamini-Hochberg correction. Figure 1 presents the full correlation matrix.

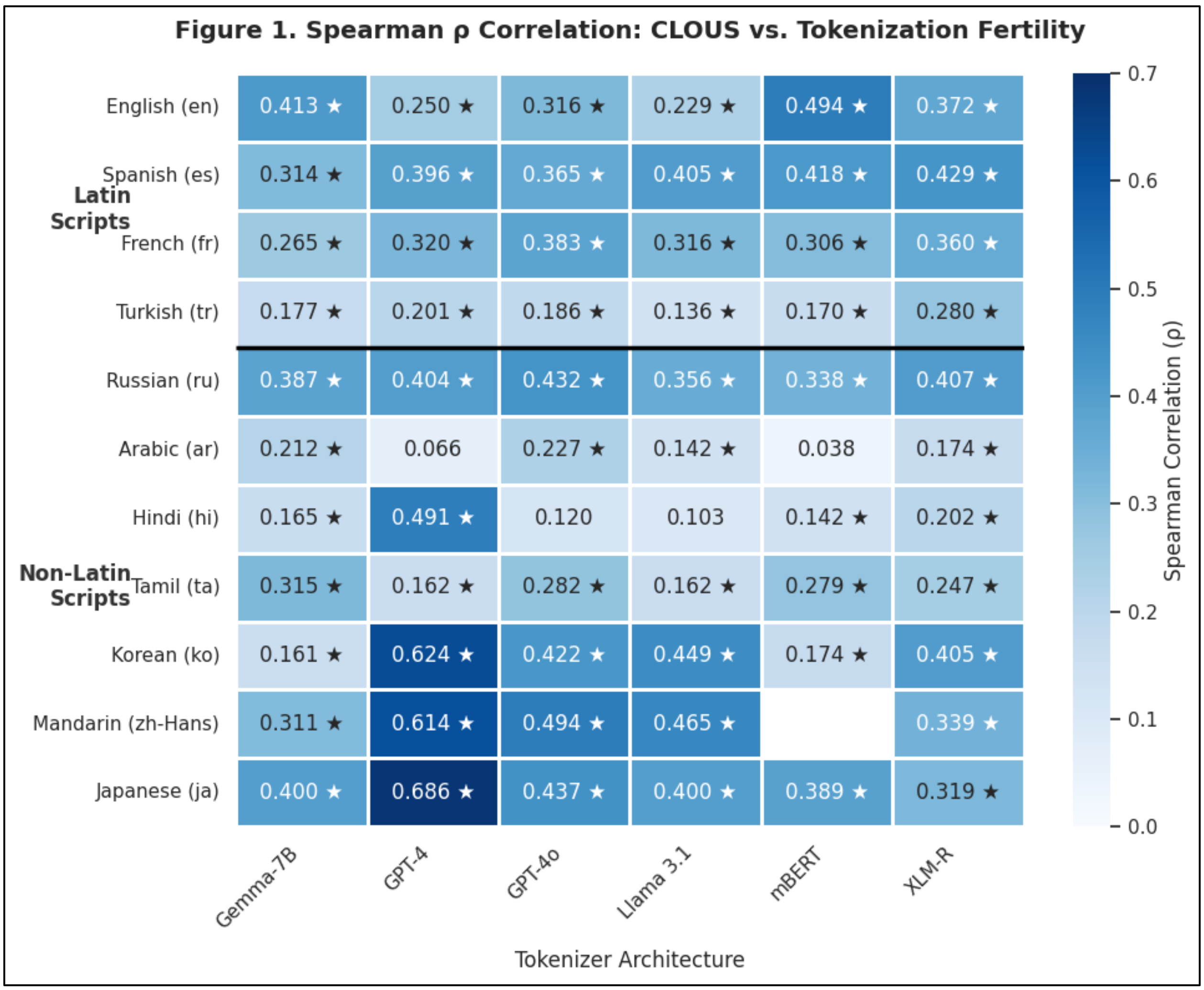

Figure 1. Spearman ρ between CLOUS scores and tokenization fertility across all language-tokenizer combinations. The mBERT × zh-Hans cell is excluded (see Appendix B).

The correlation matrix shows a consistent pattern: non-Latin script languages exhibit substantially stronger CLOUS-fertility associations than Latin-script languages. The strongest associations were observed for Japanese (GPT-4: ρ = 0.686), Korean (GPT-4: ρ = 0.624), and Mandarin Chinese (GPT-4: ρ = 0.614). Latin-script languages produced weaker but generally significant associations (ρ ≈ 0.17–0.43). Script family appears to be the primary driver of this pattern: entities whose characters are orthographically distant from the tokenizer's dominant training script tend to show greater alignment between surprisal and fragmentation severity. To confirm the robustness of these findings, we computed Kendall's τ in parallel with Spearman ρ. Both metrics demonstrated consistent directional agreement across all significant cells, with full values available in the open-source repository.

4.2 CLOUS vs. Baseline Metrics

Table 3 compares the median Spearman ρ between fertility and each of the three metrics (CLOUS, byte length, and character count) averaged across tokenizers, grouped by script family.

Script Family	CLOUS (ρ median)	Byte Length (ρ median)	Char Count (ρ median)
Latin	0.36	-0.23	-0.28
Latin (extended)	0.18	-0.30	-0.36
Cyrillic	0.40	-0.24	-0.22
Arabic	0.16	-0.32	-0.32
Devanagari	0.15	-0.43	-0.43
Tamil	0.26	-0.52	-0.52
Hangul	0.41	-0.38	-0.38
Logographic	0.47	-0.12	-0.12
Mixed-script	0.40	-0.17	-0.17

Table 3. Exact median Spearman ρ with tokenization fertility for CLOUS vs. two length-based baselines, grouped by script family across 65 valid language-tokenizer combinations.

CLOUS consistently shows positive associative alignment with fertility across all script families. Notably, both length-based baselines exhibit negative correlations with fertility. This negative association is expected: as string length increases, tokenizers have more opportunities to utilize longer multi-character merges, thereby decreasing the overall tokens-per-character ratio. Because CLOUS maintains a positive correlation, it demonstrates that character-level surprisal successfully captures information about the distributional mismatch between entity orthography and tokenizer training data, acting as a distinct signal rather than a simple proxy for string length. The divergence is most pronounced for Tamil, where CLOUS maintains a positive median correlation of +0.26 while byte length produces −0.52, a gap of 0.78 ρ units, reflecting that Tamil script characters occupy 3 bytes each in UTF-8, making longer strings paradoxically easier to tokenize as the tokenizer accumulates enough context for multi-character merges.

Figure 2 displays scatter plots of CLOUS scores against fertility for all six tokenizers, with points colored by script family.

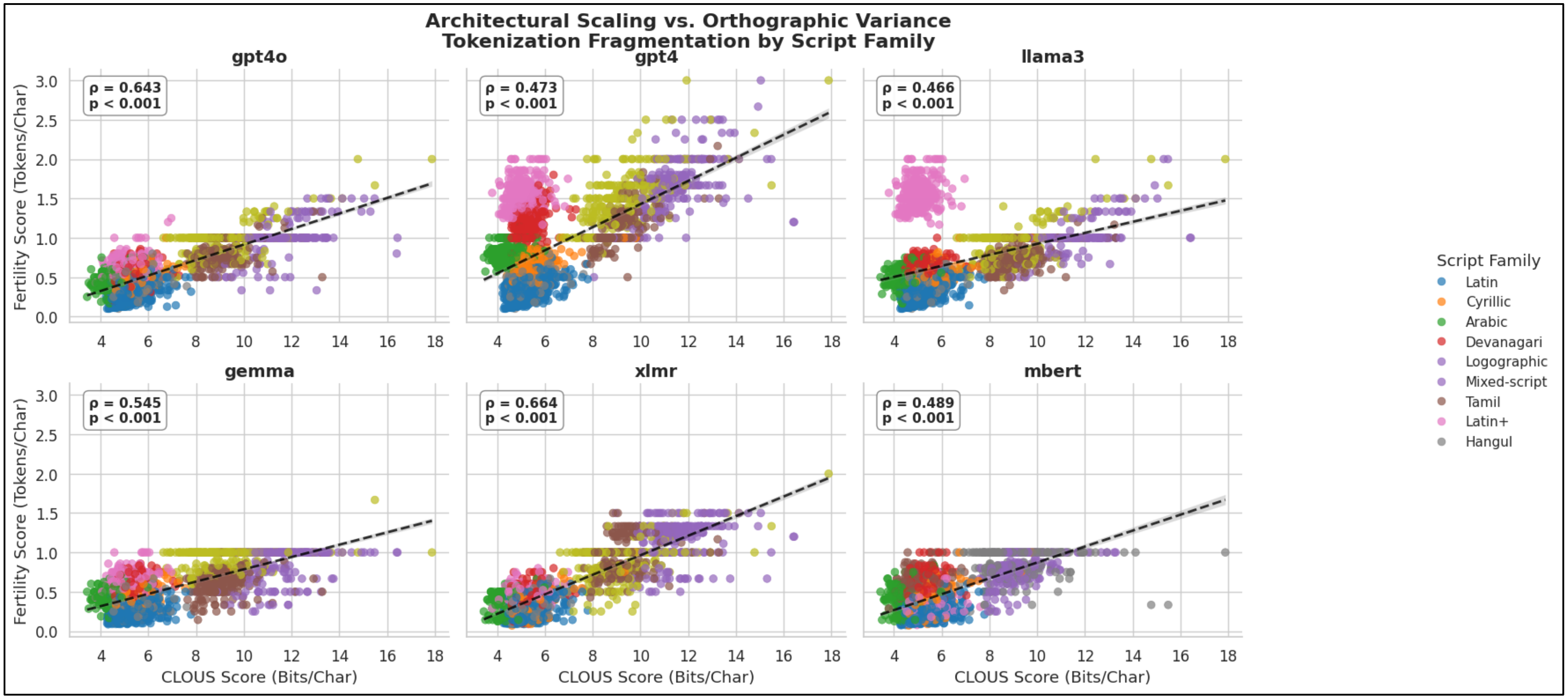

Figure 2. CLOUS score versus tokenization fertility across six tokenizers. Points are colored by script family. The GPT-4 tokenizer (100k vocabulary) produces stronger correlations for non-Latin scripts than GPT-4o (200k vocabulary) due to a ceiling effect discussed in Section 4.2.

4.3 Vocabulary Scaling Does Not Uniformly Reduce Fragmentation

Comparing correlations across tokenizer columns in Figure 1 shows that vocabulary scaling does not produce a monotonic improvement in CLOUS-fertility alignment. The GPT-4 tokenizer (cl100k_base, 100,256 tokens) shows stronger associations for Japanese, Korean, and Mandarin than GPT-4o (o200k_base, 200,000 tokens) for the same scripts. This pattern is consistent with a ceiling effect: GPT-4's smaller vocabulary fragments non-Latin scripts more severely and more uniformly, producing higher absolute fertility values but also greater entity-level variance which CLOUS can track. GPT-4o's larger vocabulary reduces absolute fertility for non-Latin scripts but compresses the variance available for correlation. This observation does not imply that smaller vocabularies are preferable; it reflects a measurement artifact of using rank correlation as the evaluation criterion.

4.4 High-CLOUS Entities: Three Orthographic Typologies

Table 4 lists the fifteen country names with the highest mean CLOUS scores across all eleven languages.

ISO	Country Name	Mean CLOUS	Typology	Fert. (mBERT)	Fert. (GPT-4o)	Δ (Fertility Change)
GU	Guam	7.586	Phonetic Loanword	0.63	0.77	+0.14
IM	Isle of Man	7.582	Logographic Compound	0.56	0.62	+0.06
NF	Norfolk Island	7.488	Logographic Compound	0.49	0.65	+0.16
CD	Congo, Dem. Rep.	7.451	Punctuation Anomaly	0.54	0.56	+0.02
HK	Hong Kong	7.429	Logographic Compound	0.55	0.46	-0.09
CG	Congo	7.428	Logographic Compound	0.52	0.54	+0.02
MM	Myanmar	7.407	Phonetic Loanword	0.54	0.64	+0.10
FJ	Fiji	7.341	Phonetic Loanword	0.64	0.75	+0.11
KP	Korea, DPR	7.225	Logographic Compound	0.52	0.58	+0.06
VI	Virgin Islands, U.S.	7.212	Punctuation Anomaly	0.51	0.53	+0.03
CK	Cook Islands	7.128	Logographic Compound	0.49	0.56	+0.07
CI	Côte d'Ivoire	7.125	Punctuation Anomaly	0.58	0.60	+0.03
MO	Macao	7.121	Logographic Compound	0.57	0.50	-0.07
BV	Bouvet Island	7.120	Logographic Compound	0.57	0.66	+0.09
ZW	Zimbabwe	7.076	Phonetic Loanword	0.59	0.61	+0.02

Table 4. Fifteen countries with the highest mean CLOUS scores across all eleven languages.
Δ = Fertility(GPT-4o) − Fertility(mBERT). Positive values indicate GPT-4o produces higher fertility (more fragmentation) than mBERT for that entity; negative values indicate GPT-4o achieves lower fragmentation, consistent with vocabulary expansion providing coverage benefit. The majority of high-CLOUS entities show positive Δ, indicating that vocabulary scaling does not reduce (and frequently increases) fragmentation for the most orthographically unusual geographic names

Qualitative analysis of the high-CLOUS group identifies three recurring orthographic patterns:

Punctuation and Diacritic Anomalies (e.g., Côte d'Ivoire; Virgin Islands, U.S.): These entities contain non-alphabetic characters (commas, periods, apostrophes, or rare diacritics) that appear at very low relative frequency in formal news corpora. Their presence inflates mean per-character surprisal independently of the entity's length or phonological properties. The Δ fertility values near zero for these entities suggest that vocabulary scaling does not resolve punctuation-driven fragmentation.

Phonetic Loanword Clusters (e.g., Guam, Fiji, Zimbabwe): These names do not map cleanly to the phonemic inventory of non-Latin target languages. When transliterated into Arabic, Devanagari, Tamil, or Japanese, they require low-frequency characters reserved for foreign loanwords (such as Katakana extension characters in Japanese) producing elevated CLOUS scores and consistently higher fragmentation across multiple language contexts.

Logographic Compounding (e.g., Hong Kong, Isle of Man, Cook Islands): These entities, predominantly islands or territories, receive translations in East Asian logographic and syllabic scripts that compound low-frequency descriptive characters (e.g., 岛/島 for 'island,' 民主 for 'democratic'). These characters carry high individual surprisal relative to the high-frequency grammatical particles that dominate training corpora. Unlike the other typologies, Logographic Compounding entities show mixed results from vocabulary scaling. While highly prevalent entities like Hong Kong and Macao show measurable fertility improvement under GPT-4o (Δ = −0.09 and −0.07 respectively, indicating GPT-4o requires fewer tokens per character for these entities), the majority of Logographic Compounding entities actually experience increased fragmentation under the larger vocabulary (positive Δ), suggesting that vocabulary scaling provides uneven benefit even within a single orthographic typology.

4.5 CLOUS as a High-Risk Entity Screening Tool

To evaluate CLOUS as an operational diagnostic rather than a correlational measure, we assessed its ability to identify entities at elevated fragmentation risk without model inference. Defining the top quartile of mean fertility scores as the positive class, entities most likely to incur disproportionate tokenization cost, CLOUS achieves ROC-AUC = 0.850, precision = 0.753, recall = 0.729, and F1 = 0.741 across all entity-language pairs. These results indicate that CLOUS functions as a meaningful pre-tokenization screen, correctly identifying approximately three-quarters of the highest-fragmentation entities from orthographic properties alone.

Quartile analysis reveals a pronounced threshold effect. Entities in the lowest CLOUS quartile (Q1) exhibit a mean fertility of 0.461, comparable to Q2 (0.458), while Q3 rises modestly to 0.531. The highest quartile (Q4) reaches a mean fertility of 1.021, a 121.4% increase over Q1, indicating that the fragmentation penalty is concentrated among the most orthographically unusual entities rather than distributed uniformly across the CLOUS range.

A simple linear regression (Fertility = −0.045 + 0.102 · CLOUS) yields $R^2 = 0.383$, MAE = 0.212, and RMSE = 0.308. While CLOUS alone explains approximately 38% of fertility variance, the ROC-AUC result suggests that its practical value lies primarily in rank-ordering and screening rather than precise fertility point estimation. Practitioners seeking to audit a geographic entity set for tokenization risk can use CLOUS scores to prioritize the top quartile of entities for vocabulary augmentation or alternative representation strategies before deployment.

5. Discussion

5.1 Script Family as the Primary Driver of CLOUS-Fertility Alignment

The pattern across Figure 1 is consistent with script family being the primary factor associated with CLOUS-fertility correlation strength. Tokenizers whose training data are dominated by Latin-script text will have fewer vocabulary entries covering non-Latin character sequences, regardless of total vocabulary size. This produces a systematic relationship between character-level surprisal and fragmentation severity for non-Latin-script languages that is weaker or absent for Latin-script languages where the tokenizer's training distribution more closely matches the entity's orthographic form.

We note that this interpretation is observational. While a formal moderation analysis would quantify the script family interaction, we prioritize descriptive clarity to focus on the metric's broad predictive utility. The observed correlation patterns are sufficiently robust that complex interaction modeling is deferred to future work.

5.2 Practical Utility as a Diagnostic Tool

CLOUS requires no model inference: only a character frequency corpus and a geographic entity list. This makes it applicable as a pre-training diagnostic for identifying entities likely to fragment in a target tokenizer configuration. Practitioners building domain-specific multilingual systems with a known entity set could use CLOUS scores to prioritize vocabulary augmentation candidates before training, potentially reducing fragmentation-related performance gaps for specific geographic or named entity classes.

The baseline comparison in Table 3 suggests that character-level surprisal captures information beyond raw string length, particularly for non-Latin scripts. However, the practical significance of the CLOUS advantage over length-based baselines, which ranges from 0.28 ρ units for Latin-script languages to 0.78 ρ units for Tamil, the largest divergence observed, will depend on the specific application and entity set. We do not claim that CLOUS is the optimal metric for this purpose; rather, we propose it as a principled starting point that outperforms simple length heuristics and offers an interpretable connection to the distributional properties of the tokenizer's training data.

5.3 Practical Recommendations

1. **Prioritize Entities for Augmentation:** Developers should use CLOUS to identify geographic "hotspots" in their dataset that will trigger high fragmentation. Entities identified as high-CLOUS are prime candidates for custom dictionary or vocabulary token additions.
2. **Cost-Aware Routing:** For production systems using commercial APIs, high-CLOUS names significantly increase token cost. Practitioners should route requests containing these entities to models with more efficient, script-aware tokenizers.
3. **Validate Beyond Latin Scripts:** The predictive power of CLOUS is most pronounced in non-Latin scripts. When developing for these languages, character-level surprisal is a more reliable proxy for fragmentation risk than simple length-based heuristics."

5.4 Limitations

This study has several limitations. First, character frequency profiles were derived from formal news and web text corpora; distributions from other registers may differ in ways that affect CLOUS scores. Second, the study covers eleven languages but omits several script families that would likely show high fragmentation rates, including Southeast Asian abugidas (Thai, Khmer, Burmese) and scripts used for lower-resource languages not represented in the Leipzig Corpora Collection. Third, the correlations reported are associative rather than causal: CLOUS correlates with fertility because both are functions of the same underlying distributional mismatch, but CLOUS does not cause fragmentation. Fourth, this study focuses on country names as a controlled entity class; generalization to other named entity types, such as person names, organization names, and street addresses, requires separate empirical evaluation. Fifth, Table 3 reports exact median Spearman ρ values computed across all six tokenizers per script family from the full benchmark results; per-tokenizer breakdowns are available in the repository.

6. Conclusion

We introduced CLOUS, a character-level metric based on Shannon surprisal that estimates the orthographic rarity of geographic entity names relative to language-specific character frequency distributions. Evaluated against tokenization fertility from six production tokenizers across eleven languages, CLOUS shows significant associative strength with fertility in 61 of 65 tested combinations after multiple comparisons correction. CLOUS outperforms two naive length-based baselines (byte length and character count) in correlational alignment with fertility, with the advantage increasing substantially for non-Latin scripts. Script family appears to be the primary driver of CLOUS-fertility correlation strength, with the largest effects observed for logographic and mixed-script languages.

These findings suggest that character-level surprisal, computed from a standard frequency corpus, may be useful as a lightweight pre-training diagnostic for identifying geographic entities at elevated risk of fragmentation in multilingual NLP pipelines. Beyond correlation, CLOUS demonstrates operational utility as a pre-tokenization screening tool: treating the top fertility quartile as high-fragmentation risk, CLOUS achieves ROC-AUC = 0.850 and F1 = 0.741, enabling practitioners to flag problematic entities before model inference.

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Appendix A: Repository Structure

The full pipeline and Colab notebook are available as an open-source repository at https://github.com/ric-rivera/CLOUS-tokenization-metrics. The repository structure is organized as follows:

- **raw_corpora/**: Leipzig Corpora Collection character frequency files
- **cache/**: Serialized JSON frequency maps and intermediate pipeline results (benchmark_results.csv, final_stats.csv, and clous_scores.csv)
- **CLOUS_pipeline.ipynb**: Main Google Colab notebook
- **results/**: Figure_1_CLOUS_Heatmap.png, Figure_2_Stratified_Scatter.png

Appendix B: Excluded Combination: mBERT × zh-Hans

The mBERT WordPiece tokenizer maps each Simplified Chinese character to exactly one token, producing uniform fertility = 1.0 for all entities in this language. This eliminates the variance required for Spearman rank correlation. Table B1 illustrates the behavior.

ISO	Country (zh-Hans)	Characters	mBERT Tokens	Token Count	Fertility
IS	冰岛 (Iceland)	2	[冰, 岛]	2	1.00
UZ	乌兹别克斯坦 (Uzbekistan)	6	[乌, 兹, 别, 克, 斯, 坦]	6	1.00
DJ	吉布提 (Djibouti)	3	[吉, 布, 提]	3	1.00

Table B1. mBERT fertility for all Simplified Chinese entities is uniformly 1.0, reflecting character-level WordPiece tokenization. The SciPy statistical backend correctly raises a ConstantInputWarning for this combination, confirming zero variance. This combination is excluded from all Spearman analyses.